TapRunner Tutorial

Running a test station with TapRunner 1.3.5, using PowerON.TapPlan as the worked example

New Eagle · October 2026

This tutorial walks an operator through a complete test with TapRunner: loading a test plan, checking its parameters, running it, reading the result, and finding out why a run failed. Every screen shown was captured from TapRunner 1.3.5 with **C:\OpenTAP\PowerON.TapPlan** loaded, so the parameter names match what you will see on the station.

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1. What TapRunner does

TapRunner is a simple front end for OpenTAP test plans. It runs the OpenTAP command-line program (tap.exe) for you, so an operator never has to type a command or open the OpenTAP Editor. It shows:

- the plan's operator-editable parameters (serial number, part number and so on);
- six status lights that show whether the run is ready, running, passing or failing;
- a live list of every step as it finishes, and a separate list of only the steps that failed, with their limits.

TapRunner only runs plans; you still build and change plans in the OpenTAP Editor.

2. Before you start (one-time setup)

Install TapRunner

1. Run **TapRunnerSetup.msi**. It installs for the current Windows user only, so no administrator rights are needed.
2. The installer also installs the **TapRunner Status** plugin into OpenTAP. This is what reports live status back to TapRunner.
3. Start TapRunner from the Start menu.
4. To check that OpenTAP has the right plugin, open **C:\OpenTAP\Packages\TapRunner.StatusListener\package.xml** in Notepad: its Version should match TapRunner's (1.3.5 or later). If it is older, see section 9.

Turn on the TapRunner Status result listener

In the OpenTAP Editor, open **Settings › Results** and make sure **TapRunner Status** is in the list and enabled. Without it, the test still runs, but the status lights and step list stay blank. When it finishes, TapRunner reports "Process exited without reporting a result".

Check the station settings

Click **Settings…** on the main window and check the General tab (Figure 1). Most stations only need these set once:

Setting	Set it to
OpenTAP executable	The station's tap.exe, normally C:\OpenTAP\tap.exe.
OpenTAP profile	The bench profile for this station (the list shows the profiles found under Settings\Bench).
SQLite results database	The database file the test results are stored in. TapRunner saves this into OpenTAP's own Settings\Results.xml.
Test results viewer	The program the **View Results** button opens that database in, normally C:\OpenTAP\TestRunViewer\TestRunViewerSqlite.exe.
Working directory	The folder tap.exe runs in, normally C:\OpenTAP.
Power On / Power Off plan	The plans run by the Power On and Power Off buttons. PowerON.TapPlan and PowerOFF.TapPlan by default.
Station name	The name shown next to the operator. Defaults to the PC's name.

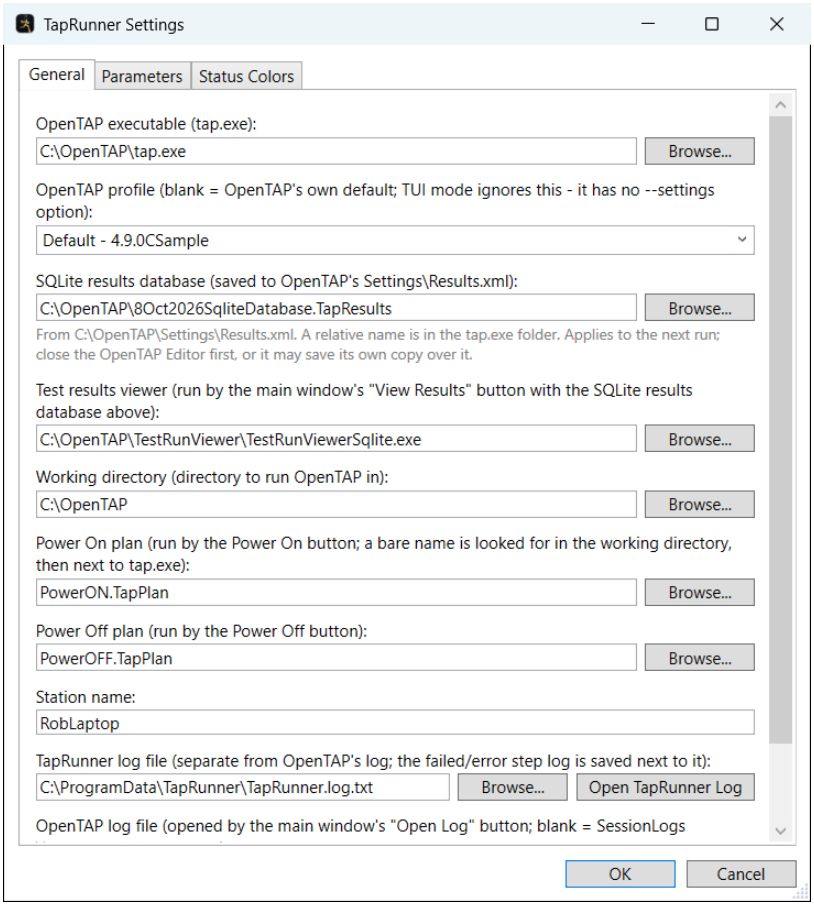


Figure 1 – Settings, General tab

3. A tour of the main window

Figure 2 shows TapRunner as it opens on the station, with PowerON.TapPlan already loaded. The numbers match the table below.

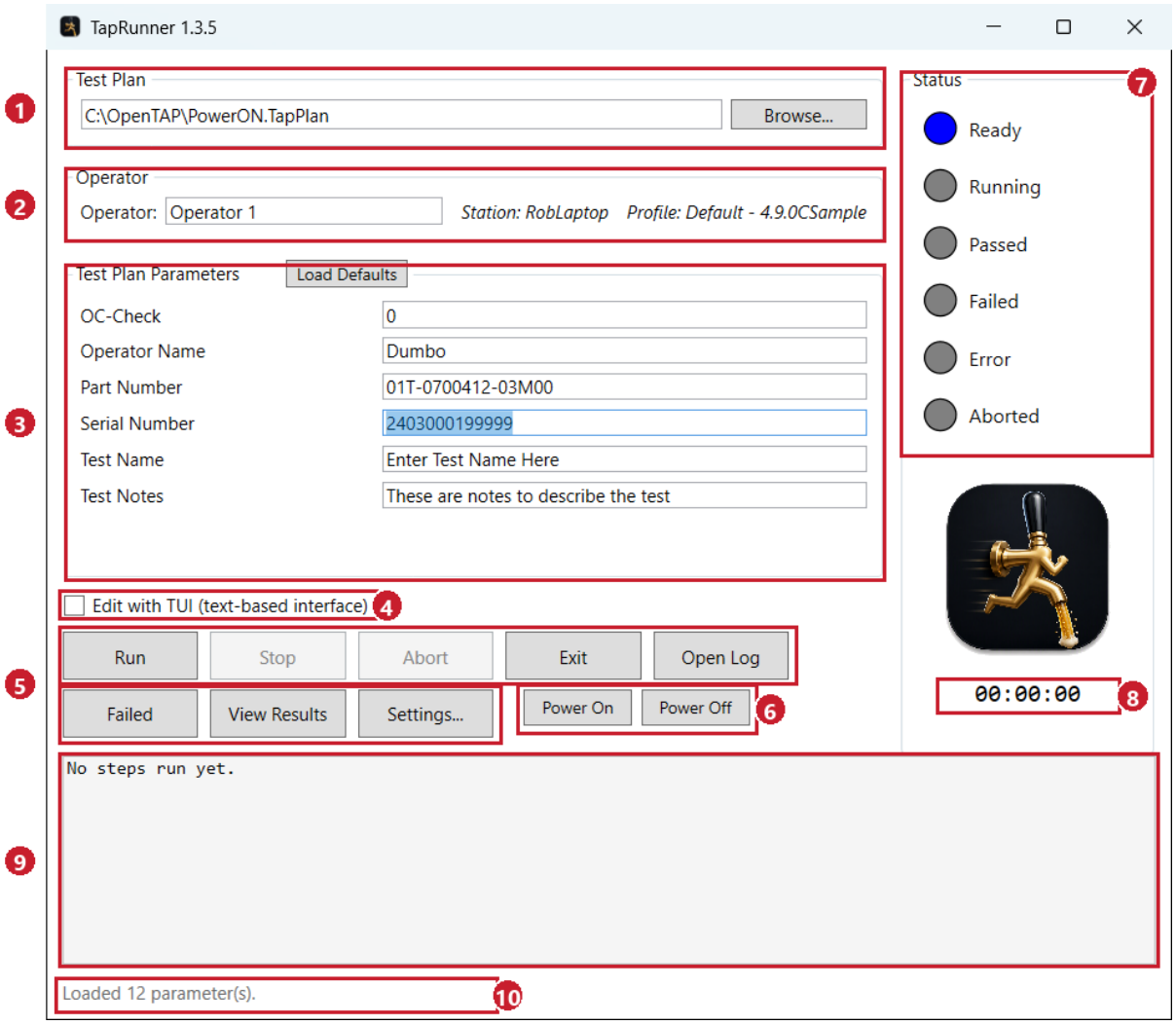


Figure 2 – The main window with PowerON.TapPlan loaded

#	Area	What it is for
1	Test Plan	The plan that will run. TapRunner reopens the last plan used; **Browse…** picks another.
2	Operator	Your name, plus the station name and OpenTAP profile from Settings.
3	Test Plan Parameters	The plan's operator-editable values. For PowerON.TapPlan: OC-Check, Operator Name, Part Number, Serial Number, Test Name and Test Notes. **Load Defaults** puts back the saved values from Settings.
4	Edit with TUI	Opens the plan in OpenTAP's text-based interface instead of running it. The Run button changes to **Edit**.

#	Area	What it is for
5	Buttons	**Run**, **Stop**, **Abort**, **Exit**, **Open Log** (OpenTAP's log), **Failed** (failed steps), **View Results** (the results database in the test results viewer) and **Settings…**.
6	Power On / Power Off	Run the station's power-on or power-off plan in one click.
7	Status lights	Ready, Running, Passed, Failed, Error and Aborted – see section 5.
8	Run time	How long the current or last run took.
9	Step history	Every step as it finishes, with its verdict and results. Holds the last 1,000 steps and can be scrolled.
10	Status line	What TapRunner is doing now, e.g. the step that is running.

4. Tutorial: run PowerON.TapPlan

Step 1 – Load the plan

If the Test Plan box does not already show C:\OpenTAP\PowerON.TapPlan, click **Browse…** and open it. TapRunner reads the plan's parameters and lists them in alphabetical order. The status line then shows how many were loaded ("Loaded 12 parameter(s)." for PowerON.TapPlan, of which six are shown and six are hidden – see section 7).

Step 2 – Check who and where

Type your name in **Operator**. Check that the station and profile shown beside it are right for this bench.

Step 3 – Fill in the parameters

The parameter values start from the defaults saved in Settings (or the plan's own defaults). For each unit:

1. **Serial Number** is already selected when the plan loads, so you can scan the unit's barcode straight in.
2. Check **Part Number** and **OC-Check** match the unit on the bench.
3. Change **Test Name** and **Test Notes** if your process asks for them.

> Values you type here apply to this run only. To change what a parameter starts at every time, change it on the Parameters tab in Settings (section 7). **Load Defaults** puts every parameter back to its saved value.

Step 4 – Run it

Click **Run**. TapRunner starts tap.exe, and its console window opens briefly before TapRunner comes back to the front. While the plan runs:

- the **Running** light is on and the run-time clock counts up;
- **Passed** flashes for as long as every step so far has passed;

- each step is added to the step history as it finishes, and the status line shows the step running now;
- Run, Browse, Settings and the Power buttons are disabled; Stop and Abort are enabled.

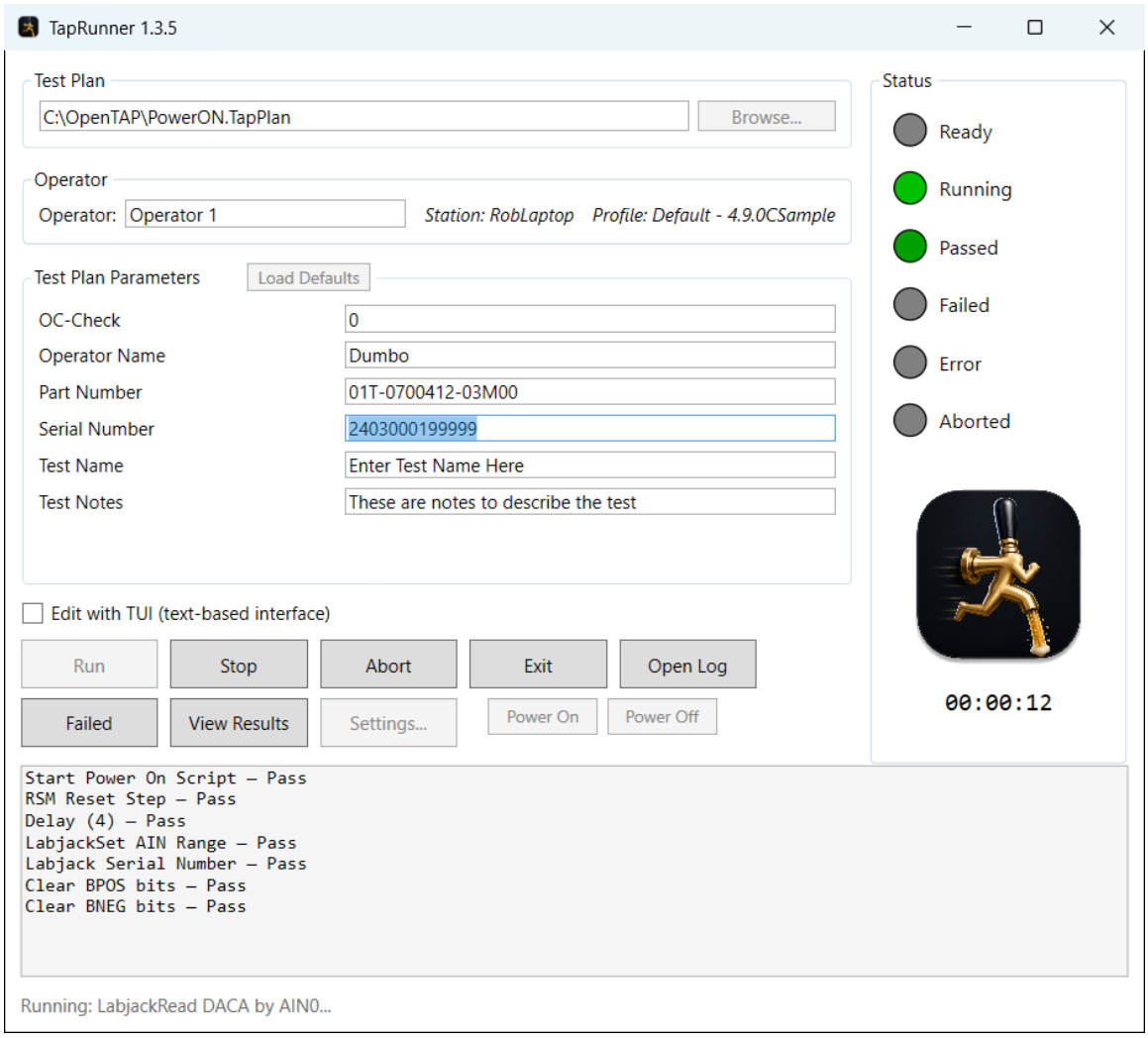


Figure 3 – PowerON.TapPlan running – every step so far has passed

Step 5 – Read the result

When the plan finishes, the light for its final verdict stays on, the clock stops, and the status line says "Run finished: Passed." (or Failed, Error, Aborted). Ready lights again and the buttons are re-enabled for the next unit.

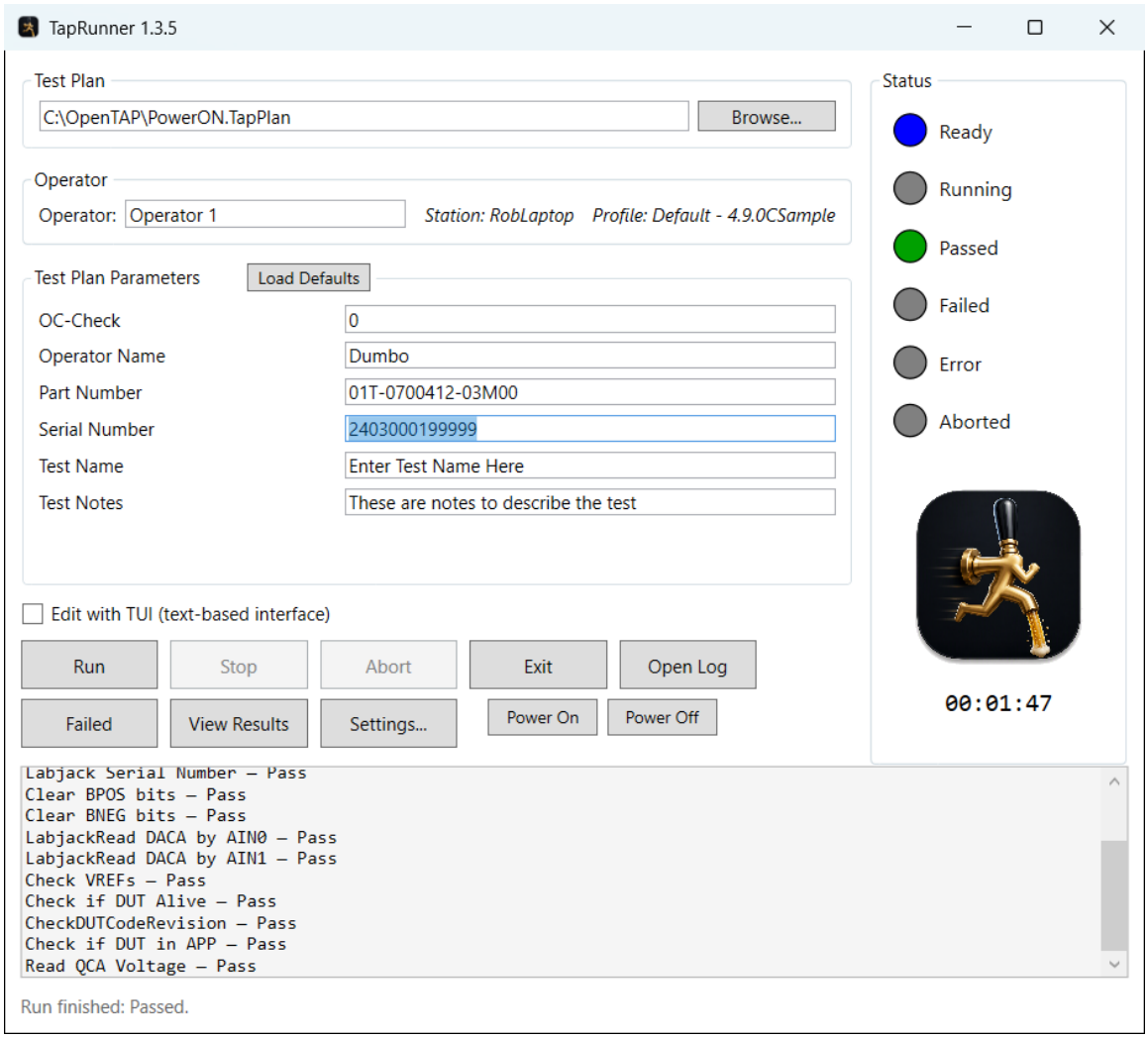


Figure 4 – A finished run with a Passed verdict

Shortcut – the Power On and Power Off buttons

Power On runs the station's power-on plan (PowerON.TapPlan by default) straight away, using its saved default parameter values. You don't need to load the plan first. The plan shown in the main window stays loaded. **Power Off** does the same with PowerOFF.TapPlan. A plan name without a folder is looked for in the working directory first, then in the tap.exe folder.

5. When a test fails

If a step fails, the **Failed** light comes on and flashes, and the step history shows which step failed and the results it published. In Figure 5, three PowerON.TapPlan measurements failed – a VREFB reading, the code version check and the QCA voltage – so the run finished Failed.

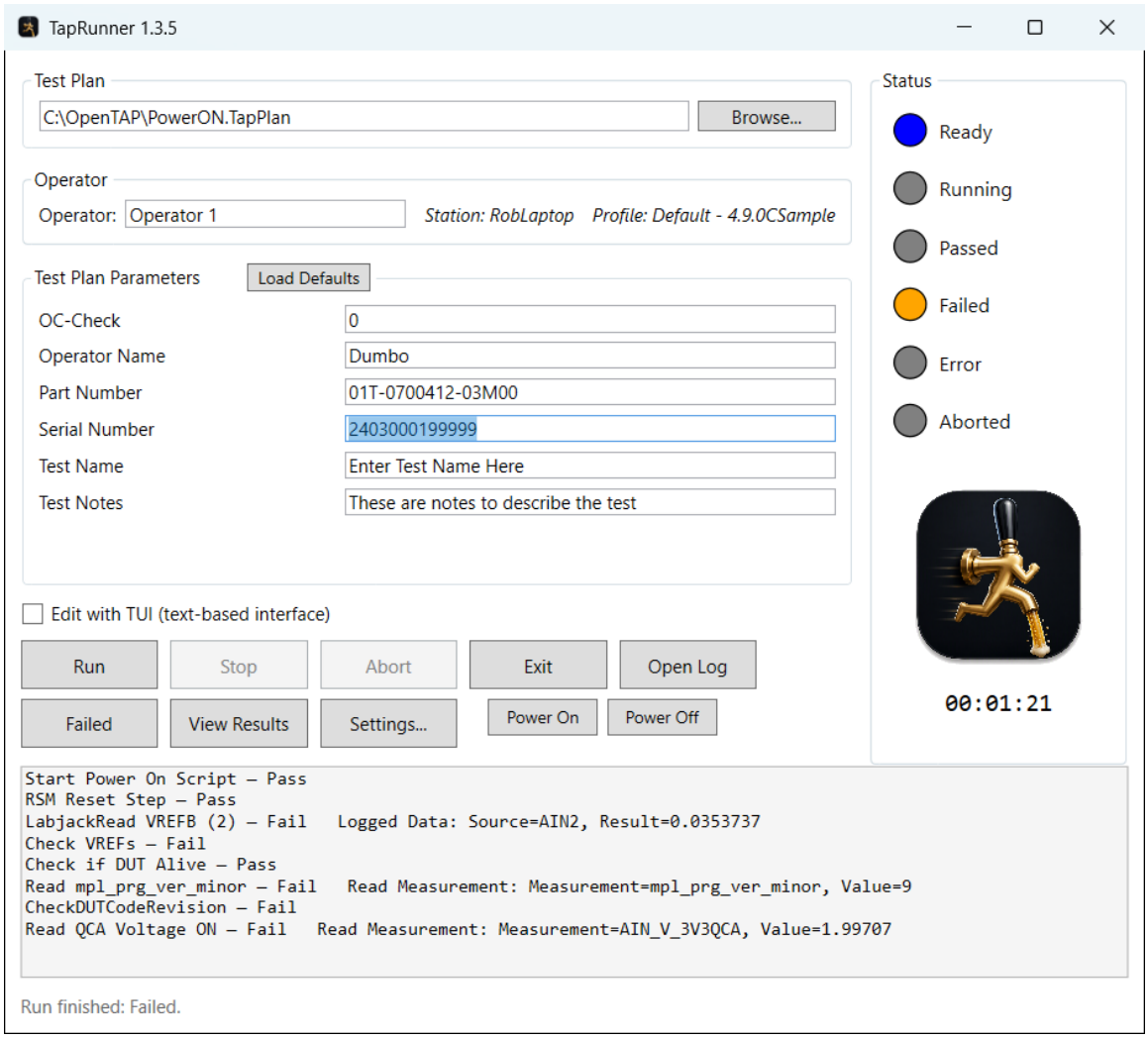


Figure 5 – A PowerON.TapPlan run with three failed measurements

Click **Failed** to see just the failed steps, with more detail than the step history (Figure 6). Each run's failures are headed with the plan name and the time the run started, and each failed measurement shows its **Limits** above the value it measured. Click **Refresh** to pick up new failures while a run is still going.

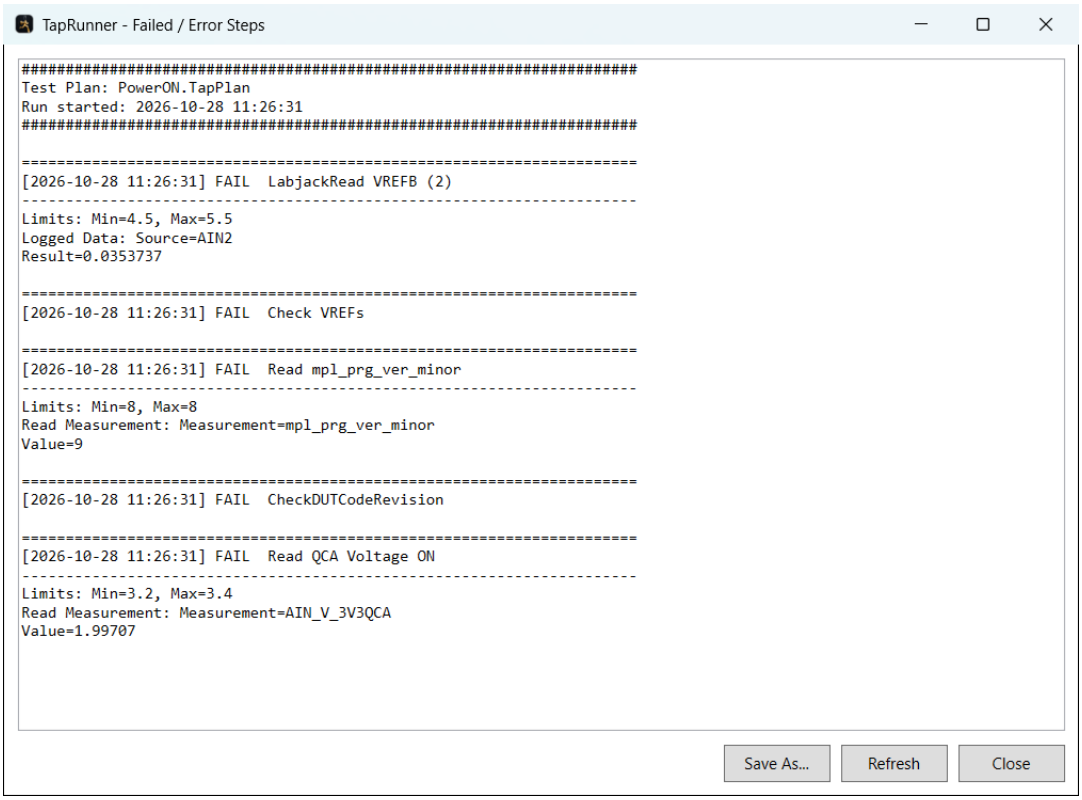


Figure 6 – The Failed / Error Steps window, with each step's limits

Reading the limits

The Limits line shows the minimum and maximum set on that step in the test plan – the same values as the step's Limits settings in the OpenTAP Editor. Compare them with the value below to see how far out the unit was. In Figure 6:

- **LabjackRead VREFB (2)** – Limits: Min=4.5, Max=5.5, but read 0.0353737: close to zero, far outside the range rather than just slightly off.
- **Read mpl_prg_ver_minor** – Limits: Min=8, Max=8, so the plan expects exactly 8, but the unit reported 9.
- **Read QCA Voltage ON** – Limits: Min=3.2, Max=3.4, but read 1.99707: well below the range.

Some entries have no Limits line. **Check VREFs** and **CheckDUTCodeRevision** are groups of steps: they fail because a step inside them failed, and have no limits of their own – look at the entry just above them. If a step's limit check is turned off in the plan, its limits are shown with "(limit check off)" after them.

> Limits need the TapRunner Status plugin from TapRunner 1.3.5 or later in OpenTAP (section 2). With an older plugin, the failure log has the measured values but no Limits lines.

Click **Save As…** to save a copy of the whole failure log to a file you choose, for example to attach to a defect report or send to an engineer.

What the status lights mean

Light	During a run	After a run
Ready	Off.	On – ready for the next run.
Running	On (flashing by default).	Off.
Passed	Flashes while every step so far has passed.	On – the plan passed.
Failed	Flashes while a step failure stands. If a parent step clears the failure (for example a retry that passes), it goes **dim** and Passed flashes again.	On – the plan failed. Dim – the plan passed, but only after a failure was cleared.
Error	Same as Failed, for steps that end in Error.	On – the plan ended in Error, including when tap.exe stopped without reporting a result.
Aborted	Off.	On – the run was stopped with Stop or Abort.

6. Stopping a run

Button	What it does
Stop	Asks OpenTAP to stop cleanly. The current step finishes and any clean-up steps run, then the run ends as Aborted. Use this one normally.
Abort	Ends tap.exe immediately. Clean-up steps do **not** run, so the hardware may be left powered or connected. Use it only when Stop doesn't work.
Exit	Closes TapRunner. If a test is running, it asks first and aborts the run if you say yes.

7. Setting up parameters for the station

Default parameter values

The **Parameters** tab in Settings (Figure 7) holds a default value for each parameter name. These are filled in every time a plan with that parameter loads, and again when you click Load Defaults. A value here always wins over the plan's own default; leave it blank to use the plan's. **Copy From Current Plan** fills the list from what is in the main window right now. That is a quick way to save a set of values you have just typed in for PowerON.TapPlan.

Hiding parameters

Tick **Hidden** next to any parameter the operator should not see. PowerON.TapPlan has six parameters that are set once per station and hidden this way: AppMajor, AppMinor, AppSub, PlatformMajor, PlatformMinor and PlatformSub. Hidden parameters are still sent to the test with their saved value. Tick **Show hidden parameters greyed out** to keep them in the list, dimmed but still editable (Figure 8).

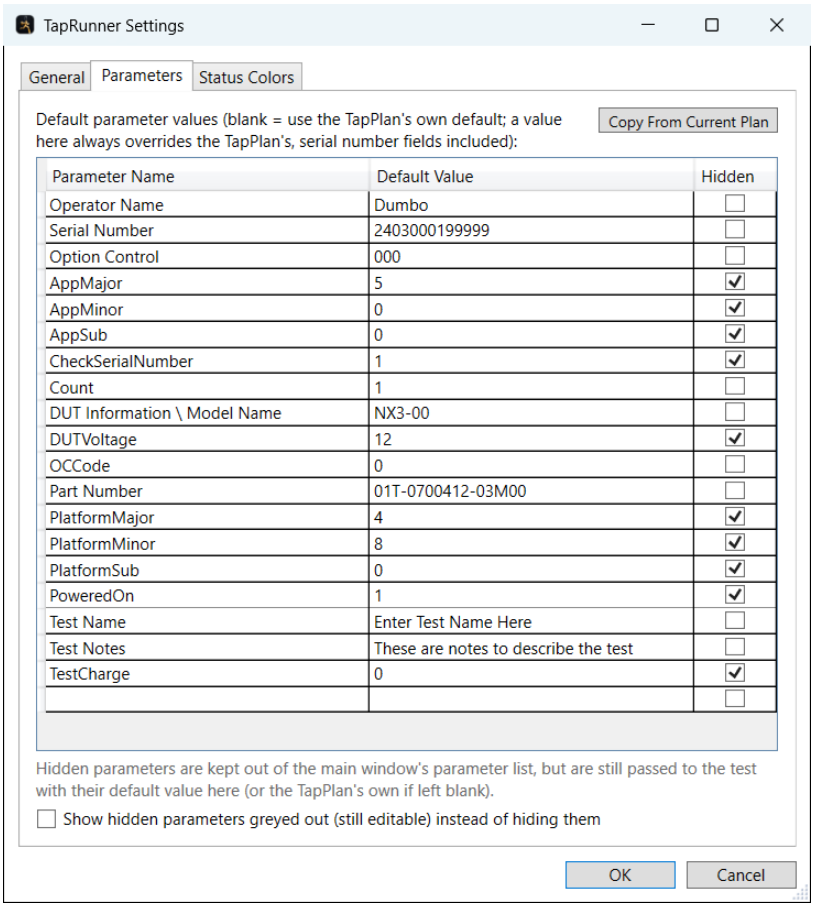


Figure 7 – Settings, Parameters tab, with the Hidden column

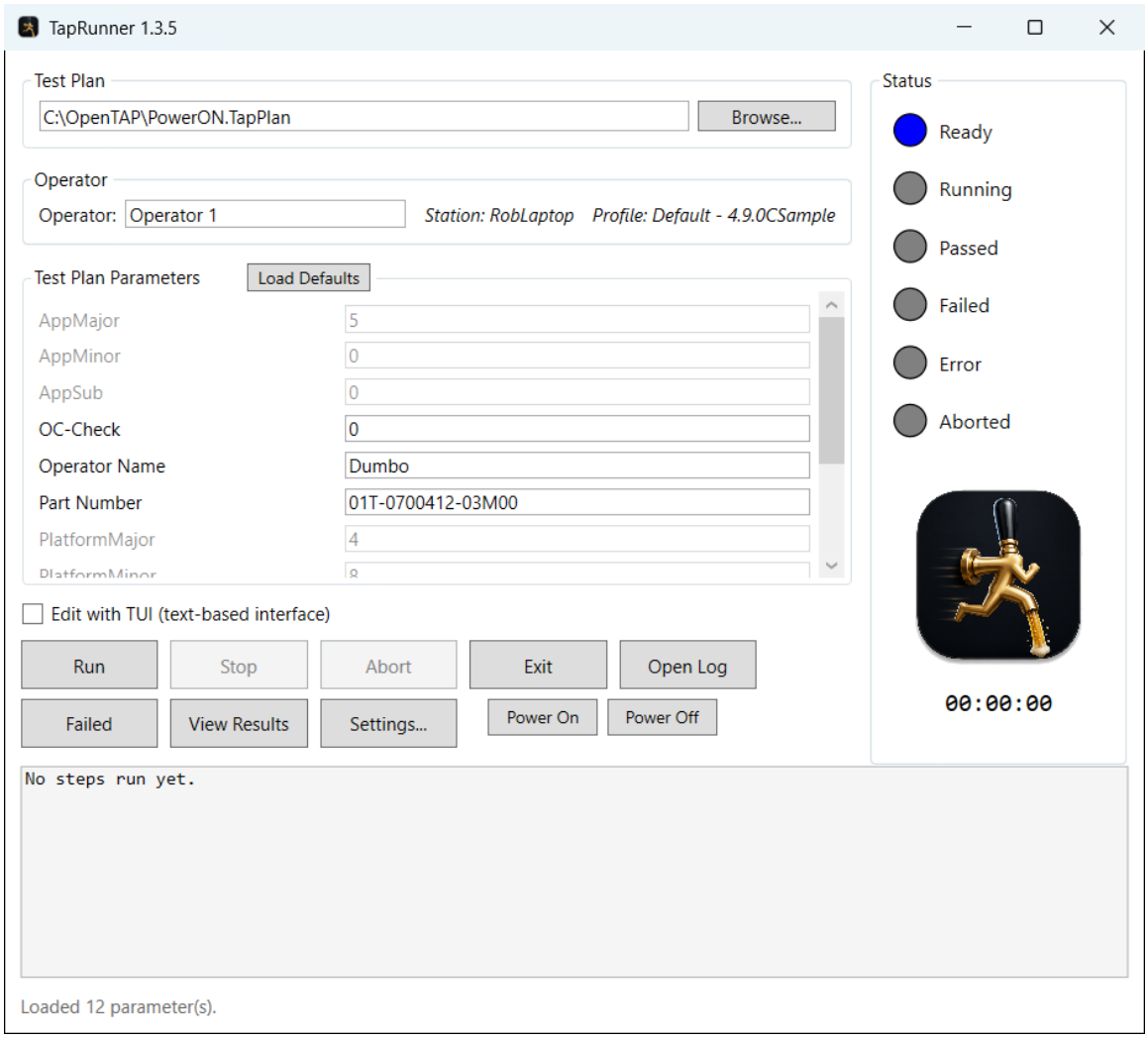


Figure 8 – PowerON.TapPlan with its hidden parameters shown greyed out

Status colours

The Status Colors tab sets the colour of each light, and whether it flashes when fully on. Enter a colour name or a #RRGGBB value.



Figure 9 – Settings, Status Colors tab

8. Logs and stored results

Where	What you get
Open Log (main window)	OpenTAP's own log of the latest session. It's the most detailed record of what OpenTAP did, including licence and instrument errors.
Failed (main window)	TapRunner's failure log: only the steps that ended Fail or Error, grouped by run, with each step's limits. Save As… saves a copy.
View Results (main window)	The SQLite results database your results are stored in, opened in the test results viewer (TestRunViewer) set in Settings, for browsing past runs.
Settings › Open TapRunner Log	TapRunner's own log: every status message and every finished step, with a timestamp, across all runs.

By default the failure log keeps building up across runs. Tick **Clear the failure log at the start of each run** in Settings if you only want the current run's failures.

9. Quick troubleshooting

You see	Try this
The lights never change, then "Process exited without reporting a result"	Enable TapRunner Status under Settings › Results in OpenTAP (section 2). If it is enabled, click Open Log and look for a licence or start-up error.
"The configured OpenTAP executable was not found"	Set the OpenTAP executable under Settings.
A parameter you expect is missing from the list	It may be ticked Hidden on the Parameters tab in Settings.
Results are going to the wrong database	Check SQLite results database under Settings. Close the OpenTAP Editor before changing it, or the Editor may save its own setting over it.
View Results says the viewer or database was not found	Set Test results viewer under Settings. If the database is new, run a test first – it is created by the first run that stores results.
The failure log has no Limits line for a step	If it's a group step (like Check VREFs), that's normal – see section 5. Otherwise the TapRunner Status plugin in OpenTAP is probably older than 1.3.5: check its Version in C:\OpenTAP\Packages\TapRunner.StatusListener\package.xml and run TapRunnerSetup.msi again. To install the plugin by hand, run tap package install from the C:\OpenTAP folder – from any other folder it can fail.
Power On says the plan was not found	Set the full path to the plan under Settings › Power On plan.